

How Properties Emerge*

Paul Humphreys†‡

Department of Philosophy, University of Virginia

A framework for representing a specific kind of emergent property instance is given. A solution to a generalized version of the exclusion argument is then provided and it is shown that upwards and downwards causation is unproblematical for that kind of emergence. One real example of this kind of emergence is briefly described and the suggestion made that emergence may be more common than current opinions allow.

1. Introduction. Lurking in the shadows of contemporary philosophy of mind is an argument widely believed to produce serious problems for mental causation. This argument has various versions, but one particularly stark formulation is this:

- “(1) If an event x is causally sufficient for an event y , then no event x^* distinct from x is causally relevant to y (*exclusion*).
- (2) For every physical event y , some physical event x is causally sufficient for y (*physical determinism*).
- (3) For every physical event x and mental event x^* , x is distinct from x^* (*dualism*).

*Received March 1996; revised September 1996.

†Send reprint requests to the author. Corcoran Department of Philosophy, 521 Cabell Hall. University of Virginia, Charlottesville. VA 22903.

‡Previous versions of this paper were read at Virginia Polytechnic Institute, Duke University. The University of Virginia, the University of Pittsburgh, the British Society for the Philosophy of Science, and an IUHPS meeting in Warsaw. Comments and suggestions from those audiences and the two anonymous referees from *Philosophy of Science* were very helpful in improving the paper. I am also grateful for conversations and correspondence with Robert Almeder, James Bogen, Richard Burian, John Forge, David Henderson, Martin Jones, Jaegwon Kim, James Klagge, Ken Olson, Fritz Rohrlich, and Abner Shimony. Research for this paper was conducted partly under NSF grant SBR-9311982 and the support is gratefully acknowledged.

Philosophy of Science, 64 (March 1997) pp. 1–17. 0031-8248/97/6401-0001\$2.00
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- (4) So: for every physical event y , no mental event x^* is causally relevant to y (*epiphenomenalism*).” (Yablo 1992, 247–248).¹

This *exclusion argument*, as it is usually called, has devastating consequences for any position that considers mental properties to be real, including those non-reductive views that suppose mental properties to supervene upon physical properties. For if mental properties are causally impotent vis-a-vis physical properties, the traditional worry about epiphenomenalism confronts us: What is the point of having them in our ontology if they are idle? Abstract objects escape this worry, for we do not expect them to do causal work, but mental properties are retained in part because we believe them to affect the course of the world. If the exclusion argument is sound, then ratiocination, qualia, and the hopes and fears of mankind are simply smoke on the fire of brain processes.

This is bad enough, but there is a second argument, devised by Jaegwon Kim, that in conjunction with the exclusion argument seems to render nonreductive physicalism not merely uncomfortable but untenable, for it has as a conclusion that nonreductive physicalism is committed to the view that some mental properties must cause physical properties. Kim’s argument, which I shall call *the downwards causation argument* was originally levelled against both nonreductive and emergentist approaches:

But why are emergentism and nonreductive physicalism committed to downward causation, causation from the mental to the physical? Here is a brief argument that shows why. At this point we know that, on emergentism, mental properties must have novel causal powers. Now, these powers must manifest themselves by causing either physical properties or other mental properties. If the former, that already is downward causation. Assume then that mental property M causes another mental property M^* . I shall show that this is possible only if M causes some physical property. Notice first that M^* is an emergent; this means that M^* is instantiated on a given occasion only because a certain physical property P^* , its emergence base, is instantiated on that occasion. In view of M^* ’s emergent dependence on P^* , then, what are we to think of its causal dependence on M ? I believe that these two claims concerning why M^* is present on this occasion must be reconciled, and that the only viable way of accomplishing it is to suppose that M caused

1. See Yablo 1992, 247, fn. 5 for a partial list of versions of this argument that have appeared in the philosophical literature. I note here that he does not endorse the simple version of the exclusion argument because it is unsound.

M* by causing its emergence base P*. In general, the principle involved here is this: *the only way to cause an emergent property to be instantiated is by causing its emergence base property to be instantiated*. And this means that the “same-level” causation of an emergent property presupposes the downward causation of its emergent base. That briefly is why emergentism is committed to downward causation. I believe that this argument remains plausible when emergence is replaced by physical realization at appropriate places. (Kim 1992, 136)

Conjoin this second argument with the first, and you have more than mere trouble for nonreductive physicalism and emergentism, you have contradictory conclusions. (4) entails that there is no downwards causation from the mental to the physical; the downwards causation argument concludes that nonreductive physicalism and emergentism require such downwards causation.

Something must go if mental properties are to survive, and that something is both arguments. Much is wrong with the exclusion argument, but what it shares with the downward causation argument is a pinched commitment to a dualist ontology, a laudable but usually unargued allegiance to the causal closure of the physical realm, and (nowadays) the idea that supervenience is the right way to represent the relation between the lower and higher levels of the world’s ontology. Each of these is popular and each is wrong.

2. A Wider Perspective. The first thing to note about these arguments is that they are extremely general—they do not seem to rely on anything that is characteristic of mental properties, such as intentionality, lack of spatial location, having semantic content, and so on, and indeed it is often mentioned in passing that the arguments can be generalized to apply to a hierarchically ordered set of properties, each level of which is distinct from every other level.² If the exclusion argument does generalize to such hierarchies, and if, for example, chemical and biological events occupy higher levels than do physical events, then no chemical or biological event could ever causally influence a physical event, and if both arguments so generalize, then nonreductive physicalism leads to inconsistencies when applied to the general realm of the natural sciences too.

The situation is in fact more extreme than this, because most of our physical ontology lies above the most fundamental level, and in consequence only the most basic physical properties can be causally effi-

2. See, e.g., Yablo 1992, 247, fn 5.

cacious if these arguments are correct. Indeed, unless we have already isolated at least some of the most fundamental physical properties, every single one of our causal claims within contemporary physics is false and consequently there are at present no true physical explanations that are grounded in causes.

All of these are, of course, surprising and unwelcome consequences. We tend to believe that current elementary particle physics does reliably describe the formative influences on our world, that chemical corrosion can cause the physical failure of an aircraft wing, and that the growth of a tree can cause changes in the ground temperature beneath it. Yet our beliefs on this score may well be wrong. Perhaps the correct formulation of physicalism is a strict one: if, as many hold, everything is composed of elementary particles or fields, and the laws that govern those objects ultimately determine all else that happens, then there are fundamental physical events that are causally sufficient for aircraft wing failures, changes in soil temperatures, and all other physical events. This is a position that requires serious consideration, and indeed those versions of physicalism that require all non-physical phenomena to supervene upon physical phenomena can be easily and naturally adapted to this kind of strict position. So we need to examine in detail a generalized version of each argument to see whether this radical physicalist conclusion can be supported and whether the problems for non-reductive physicalism extend all the way down to the penultimate level.

There is a second reason for generalizing the arguments. It is preferable to examine arguments in a form in which the contextual assumptions are as transparent as possible. The relation between molecular chemistry and physics, say, is much more easily assessed than is the relation between physics and psychology because we have clearly articulated theories for the first pair, together with a reasonably clear set of constraints on the degree of reducibility of molecular chemistry to particle physics, but those relations are much murkier in the case of physics and psychology. In addition, by focusing on the lower levels of the hierarchy, we can avoid difficult problems involving the mental that are here irrelevant simply by recasting the arguments in a form that avoids reference to specifically mental properties.³

To generate the general arguments, we need a hierarchy of levels. For present purposes, I shall simply accept that this can be done in

3. This allows, of course, that the realm of the mental does generate additional peculiar problems. The exclusion and downwards causation arguments are entirely independent of those peculiarities, however, except perhaps for a prejudice against the mental. Were it not for that prejudice, the exclusion argument could be run in reverse to exclude physical causes, a feature emphasized to me by Peter Dlugos.

whatever way the reader finds most congenial. (The strata must correspond to some real differences in ontological levels rather than to a mere set of epistemic distinctions.) For concreteness, we can think in terms of this assumption: (L) There is a hierarchy of levels of properties $L_0, L_1, \dots, L_n, \dots$ of which at least one distinct level is associated with the subject matter of each special science, and L_j cannot be reduced to L_i for any $i < j$.

I shall not try here to give any additional criteria for distinguishing levels. Rather, I am simply adopting, for the purposes of the argument, the abstract assumption of both arguments that there is indeed some such hierarchy. At the very least, one would have to consider this an idealization of some kind, and we shall see that the assumption that there is a discrete hierarchy of levels is seriously misleading and probably false. It seems more likely that even if the ordering on the complexity of structures ranging from those of elementary physics to those of astrophysics and neurophysiology is discrete, the interactions between such structures will be so entangled that any separation into levels will be quite arbitrary.

3. The Generalized Exclusion Argument. As a general principle, premise (1) is false as we originally formulated it, for it asserts that, first, causal antecedents of x and, second, events causally intermediate between x and y are causally irrelevant to y , and this is obviously wrong.

So, let us use the preliminary definition:

An event z is *causally connected* to a second event x if and only if x causes z or z causes x . z is *causally disconnected* from x just in case z is not causally connected to x . The proper formulation of (1) is then:

(1') If an event x is causally sufficient for an event y , then no event x^* distinct from x and causally disconnected from x is causally relevant to y . (exclusion)⁴

To make the revised principle true requires a strict criterion of event identity so that in particular, the exact time and way in which an event occurs is crucial to that event having the identity it does. This criterion is needed to exclude cases where x is sufficient for y but x^* , which is causally disconnected from x , brings about (a somewhat earlier analogue of) y before the connecting process from x has brought about y .

4. I am here setting aside overdetermining events as genuine examples of causation. Although some discussions of the exclusion argument see acceptance of overdetermination as a way out of the difficulty, this is not a convincing move, and I shall not follow that route. Not the least reason for this is that cases of simultaneous overdetermination are exceedingly rare.

Such a strict criterion, called ‘fragility’ by David Lewis,⁵ is controversial, and rightly so, but I shall accept it here simply because its adoption avoids distracting issues and does not affect the essential features of the argument.

Suppose now that we accept principle (L) and ask whether an analogue of premise (2) is plausible for each level L. What I shall call i-determinism (which is what a generalization of (2) asserts) is different from i-closure, the thesis that all events that are causally relevant to a given i-level event are themselves i-level events: i-determinism is compatible both with upwards and with downwards causation, where an i-level event causes another i-level event through a chain involving events at other levels. To see that (2) is not true when generalized, consider (2) with ‘physical’ replaced with ‘biological’. It might now be true that for every future biological event there is some biological antecedent that guarantees it. But to assert biological determinism for every biological event in the history of the universe would be to rule out what is commonly believed, which is that biological phenomena were not always present during the development of the universe.⁶ The first biological event, under whatever criterion of ‘biological’ you subscribe to, must have had a non-biological cause. So a generalization of (2) is not plausible for any level above the most fundamental level of all, which we call the 0-level, and so we shall restrict ourselves to a formulation of (2) for that level only, the level of whatever constitutes the most fundamental physical properties, *viz.*:

(2') For every 0-level event y, some 0-level event x is causally sufficient for y. (0-level determinism)⁷

The third premise requires a criterion of distinctness of events (as does (1')). This cannot be done in terms of spatiotemporal distinctness

5. See Lewis 1986, Postscript E. Lewis rejects extreme versions of the fragility approach.

6. This temporal development gives rise to evolutionary emergence. I shall not pursue that topic here but the reader can easily develop such an account from materials in this paper.

7. Premise (2') is, on current evidence, false because fundamental physics appears to be indeterministic in certain respects. I have preserved the original form of the argument to keep things simple, but to allay worries about the truth status of (2'), one can reformulate the argument thusly: Replace “causally sufficient” in (1') and (2') by “causally complete,” “event” by “set of events,” and x by {x_i}, where “causally complete” means either that all events necessary in the circumstances for y are included in the set or that all events that are probabilistically relevant to y are in the set. See, e.g., Lewis 1986 for one account of the former, Humphreys 1989 for one account of the latter. (3), (4) remain as they are. (2') will be false if the universe had a first uncaused event, but that fact is irrelevant here.

alone, because on supervenience accounts the supervening event is spatiotemporal coincident with the subvenient event(s). So the burden of characterizing distinctiveness will have to lie on principle (L) and we shall take as a sufficient condition for two events being distinct that they occupy different levels in the hierarchy. With this understanding we have:

- (3') For every 0-level event x and every i -level event x_i^* ($i > 0$) x is distinct from x_i^* , (pluralism)

Then it follows immediately that:

- (4') For every 0-level event y , no i -level event x_i^* ($i > 0$) that is causally disconnected from every 0-level event antecedent to y is causally relevant to y .

This modified conclusion shows how nonreductive physicalism can avoid the conclusion of the simple version of the exclusion argument and hence can also avoid the overall contradiction with the conclusion of the second argument, because the conclusion (4') allows higher level events to causally affect 0-level events if the former are part of causal chains that begin and end at the 0-level.⁸ In order for us to use that possibility to argue for emergent properties, we need to address the second argument, which amongst other things, precludes higher level causal chains that do not involve 0-level events.

4. Generalizations of the Downwards Causation Argument. In looking at generalizations of the downwards causation argument, it is worthwhile to again lay out explicitly the assumptions which underlie it. The first is a supervenience assumption that permeates the contemporary literature on nonreductive physicalism and is retained by Kim for emergent properties.⁹

- (5) Every emergent property is supervenient upon some set of physical properties.

One natural generalization of this is:

- (5') Every j -level property ($j > 0$) is supervenient upon some set of i -level properties, for $i < j$.

8. More complex versions of the argument obviously allow similar possibilities for causal chains beginning and ending at higher levels than 0.

9. Kim actually examines both the realizability and the supervenience approaches. I restrict myself to the latter.

Strict physicalists might want to insist that all higher level properties supervene upon 0-level properties. However, unlike the first argument, where premise (2) was true only for 0-level properties, here we can maintain full generality. In fact, if there are emergent properties, the strict physicalist position will be false, and we shall have to leave room for some (non-emergent) properties to supervene upon j -level emergent properties but not upon 0-level properties alone. Next, the assumption explicitly cited by Kim:

- (6) The only way to cause an emergent property to be instantiated is by causing its (set of) emergence base properties to be instantiated.

Its generalization will be (assuming that supervenience is a transitive relation):

- (6') The only way to cause a j -level property to be instantiated is by causing a set of i -level properties ($i < j$), the subvenient basis, to be instantiated.

Premises (5') and (6') are closely related, for the plausibility of (6') rests on accepting something like (5'), the idea being that emergent properties cannot exist separately from whatever physical properties give rise to them.

Then we have the important condition:

- (7) A property is emergent only if it has novel causal powers¹⁰

We can retain this unchanged for the generalized argument.

5. An Emergentist Answer to the Second Argument. How can we now escape the conclusion of the downwards causation argument? We can begin by refining the event ontology used in the argument, which appeals to properties as causes. This way of speaking, about property causation, is clearly an abbreviation for an instance of one property causing an instance of another property. We shall have to resort here to a certain amount of notational clutter. This will not be pretty, but it has a certain suggestiveness that may be helpful. From here on I shall talk of i -level properties and entities, P_m^i and x_i^i , respectively, for $i \geq 0$. I call a property (entity) an i -level property (entity) if i is the first level

10. In fact, as Martin Jones pointed out to me (pers. comm.), the novelty of the causal powers seems to play no role in Kim's central argument. Even if mental properties produced familiar physical consequences that could also be brought about by physical properties, the argument would still hold. The use of novelty is primarily in characterizing the difference between emergent and nonemergent properties, for it is an essential feature of emergent properties that they be new.

at which instances of $P_m^i(x_r^i)$ occur. i -level properties may, of course, also have instances at higher levels as for example, physical properties such as mass and volume, do. We need to keep distinct i -level entities and i -level properties, for it is possible that, in general, i -level entities may possess j -level properties, for $i \neq j$. However, for simplicity in what follows we shall assume that i -level properties are instantiated by i -level entities. So we have that $P_m^i(x_r^i)(t_1)$ causes $P_n^i(x_s^i)(t_2)$ where $t_2 > t_1$ and x_r^i, x_s^i , the entities possessing the properties, may or may not be the same.¹¹ Here P_m^i is the m th i -level property; x_r^i is the r th i -level entity, and $P_m^i(x_r^i)(t_k)$ denotes the instantiation of P_m^i by x_r^i at time t_k .

Suppose now that the i -level properties constitute a set $I = P_1^i, \dots, P_n^i \dots$ ¹² and that these i -level properties are complete in the sense that I is exhaustive of all the i -level properties. Now introduce a *fusion operation* $[*]$, such that if $P_m^i(x_r^i)(t_1), P_n^i(x_s^i)(t_1)$ are i -level property instances, then $[P_m^i(x_r^i)(t_1) * P_n^i(x_s^i)(t_1)]$ is an $i + 1$ -level property instance, the result of fusing $P_m^i(x_r^i)(t_1)$ and $P_n^i(x_s^i)(t_1)$. I want to emphasize here that it is the fusion operation on the property instances that has the real importance for emergence. Usually, the fusion operation acting on objects will merely result in a simple concatenation of the objects, here represented by $[(x_r^i) + (x_s^i)]$, within which the individuals x_r^i and x_s^i retain their identities. However, as we noted earlier, for full generality we would need to allow for the possibility of new $i + 1$ -level objects.¹³ Moreover, fusion usually is not instantaneous, and we should represent that fact. So I shall represent the action of fusion by $[P_m^i(x_r^i)(t_1) * P_n^i(x_s^i)(t_1)] = [P_m^i * P_n^i][(x_r^i) + (x_s^i)](t_1)$.¹⁴ For simplicity, I shall assume here that $*$ itself is an i -level operation (i.e., that it is an operation of the same level as the property instances which it fuses.)

11. Kim asserts (1992, 123) that the emergentists of the 1920s held that no new entities emerged at new levels of the hierarchy, only new properties. Allowing that to be historically accurate, it is as well to allow, at least notationally, that we might have new entities emerging as well as properties.

12. The cardinality of this set is unrestricted—the integer subscripts are used for convenience only.

13. Supervenience advocates have also recognized the need for this. See, e.g., Kim 1988.

14. There is a notationally harmless but metaphysically important ambiguity here between fusion operations on property instances and on properties. The latter is metaphysically derivative from the former in that when $[P_m^i(x_r^i)(t_1) * P_n^i(x_s^i)(t_1)]$ exists, then there is by virtue of this an instance of a novel property, signified by $P_m^i * P_n^i$, at level $i + 1$. This disambiguation sits most happily with the position that fusion brings into being new properties, a position that seems to fit well with the idea of emergence. Those who subscribe to the view that there are eternal emergent properties that are uninstantiated prior to some time can think of $P_m^i * P_n^i$ as a mere notational device indicating a move to a previously uninstantiated property at a higher level.

By a fusion operation, I mean a real physical operation, and not a mathematical or logical operation on predicative representations of properties. That is, $*$ is neither a logical operation such as conjunction or disjunction nor a mathematical operation such as set formation.¹⁵ $*$ need not be a causal interaction, for it can represent interactions of quite different kinds.

The key feature of $[P_m^i * P_n^i][(x_r^i) + (x_s^i)](t_1')$ is that it is a unified whole in the sense that its causal effects cannot be correctly represented in terms of the separate causal effects of $P_m^i(x_r^i)(t_1)$ and of $P_n^i(x_s^i)(t_1)$. Moreover, within the fusion $[P_m^i * P_n^i][(x_r^i) + (x_s^i)](t_1')$ the original property instances $P_m^i(x_r^i)(t_1)$, $P_n^i(x_s^i)(t_1)$ no longer exist as separate entities and they do not have all of their i -level causal powers available for use at the $(i + 1)$ st level.¹⁶ Some of them, so to speak, have been 'used up' in forming the fused property instance. Hence, these i -level property instances no longer have an independent existence within the fusion. In the course of fusing they become the $i + 1$ -level property instance, rather than realizing the $i + 1$ -level property in the way that supervenience theorists allow the subvenient property instances to continue to exist at the same time as the supervenient property instance. For example, the cusped panelling and brattishing that makes the fan vaulting of the King's College chapel architecturally transcendent exists simultaneously with the supervenient aesthetic glories of that ceiling. In contrast, when emergence occurs, the lower level property instances go out of existence in producing the higher level emergent instances. This is why supervenience approaches have great difficulty in properly representing emergent effects. To see this, consider the following formulation of strong supervenience:¹⁷

A strongly supervenes upon B just in case, necessarily, for each x and each property F in A, if x has F , then there is a property G in B such that x has G and necessarily if any y has G , it has F . (Kim 1984, 165)

Now let A be the fusion $[P_m^i * P_n^i][(x_r^i) + (x_s^i)](t_1')$. Upon what can this

15. In contrast, it is standard in the literature on supervenience to construe the subvenient basis in terms of sets of properties, or in terms of a disjunctive normal form of properties, where it is assumed that it makes sense to perform logical operations on properties. These devices are inappropriate for characterizing emergent properties and are a legacy of the continuing but, in my view, fruitless attempt to reconstruct causation and associated concepts logically rather than ontologically.

16. As mentioned earlier, the objects themselves will often retain their separate identities.

17. This argument carries over, with simple modifications, to the definition of weak supervenience.

supervene? Because we are here considering only the abstract possibility of emergent features, consider a very simple world in which $P_m^i(x_i^1)(t_1)$ and $P_n^i(x_i^2)(t_1)$ are the only i -level property instances occurring at t_1 , and in which there are no i -level property instances at t_1' . Then, trivially, there is nothing at t_1' at the i -level upon which $[P_m^i * P_n^i][(x_i^1) + (x_i^2)](t_1')$ can supervene. Faced with this, the supervenience advocate could try a different strategy, one that relies on the fact that the definition of strong supervenience does not require the supervenient and subvenient instances to be simultaneous. So one could use the earlier instances $P_m^i(x_i^1)(t_1)$ and $P_n^i(x_i^2)(t_1)$ themselves as the base upon which the later $[P_m^i * P_n^i][(x_i^1) + (x_i^2)](t_1')$ supervenes. Yet once one has allowed this temporal gap, the supervenience relation is in danger of collapsing into an ordinary causal relation. In order for the base instances to (nomologically) necessitate the fusion instance, the absence of all intervening defeaters will have to be included in the subvenient base, and this will give us a base that looks very much like a Millian unconditional cause.¹⁸ Whatever the supervenience relation might be, the way it is used in nonreductive physicalism is surely not as a causal relation, because that would immediately convert nonreductive physicalism into old-fashioned epiphenomenalism.

A second reason why supervenience seems to be an inappropriate representation of certain cases of fusion is given by the physical examples in Section 6, and I thus refer the reader to that section. However, because my purpose here is not to attack supervenience, but rather to provide a solution to the problem of upwards and downwards causation, I now return to that issue.

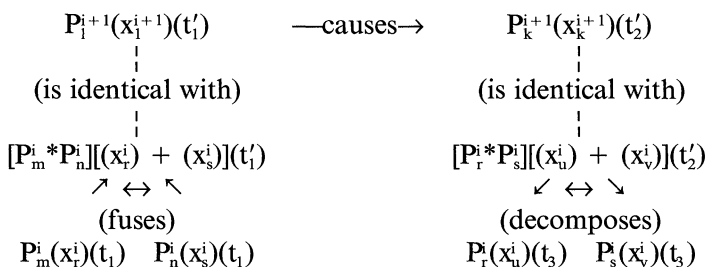
It has to be said that the unity imposed by fusion might be an illusion produced in all apparent cases by an epistemic deficit, and that when properly represented all chemical properties, for example, might be representable in terms of the separable (causal) properties of their chemical or physical constituents. But whether this can be done or not is, of course, the issue around which emergentism revolves and I shall address it explicitly in a moment in terms of some examples. What I maintain here is this: that one comprehensible version of emergentism asserts that at least some $i + 1$ -level property instances exist, that they are formed by fusion operations from i -level property instances, and that the $i + 1$ -level property instances are not supervenient upon the i -level property instances. With this representation of the emergent $i + 1$ -level property, let us add a claim that is characteristic of many versions of non-reductive physicalism, especially those motivated by multiple realizability considerations; the token identity of $i + 1$ -level prop-

18. See Humphreys 1989, Section 25, for a discussion of this condition.

erty instances and fusions of i -level property instances. That is, although we cannot identify the property P_i^{i+1} with $[P_m^i * P_n^i]$ when P_i^{i+1} is multiply realizable, we *can* identify some instances of P_i^{i+1} with some instances of $[P_m^i * P_n^i]$. It is important to remember for the purposes of the present argument that we are concerned with causal and other interactions, and not with the problem of how properties themselves are related across levels. The latter is the focus of greatest concern within nonreductive physicalism, but our problems can be solved without addressing the interlevel relationships of the properties themselves. In fact, given that the higher level properties are emergent, there is no reason to identify them with, or to reduce them to, combinations of lower level properties. Coupled with our previous reminder that it is property instances that are involved in causal relations, and not properties directly, we now have a solution to the downward causation problem for the case of emergent properties.¹⁹

Suppose that $P_i^{i+1}(x_i^{i+1})(t'_1)$ causes $P_k^{i+1}(x_k^{i+1})(t'_2)$, where both of these instances are at the $i + 1$ -level. What we have is that the i -level property instances $P_m^i(x_m^i)(t_1)$ and $P_n^i(x_n^i)(t_1)$ fuse to produce the $i + 1$ -level property instance $[P_m^i * P_n^i][(x_m^i) + (x_n^i)](t'_1)$, which is identical with $P_i^{i+1}(x_i^{i+1})(t'_1)$. This $i + 1$ -level property instance then causes the second $i + 1$ -level property instance $P_k^{i+1}(x_k^{i+1})(t'_2)$. This second $i + 1$ -level property instance, *if it is also emergent*, will be identical with, although not result from, a fusion of i -level property instances $[P_r^i * P_s^i][(x_r^i) + (x_s^i)](t'_2)$. But there is no direct causal link from the individual property instances $P_m^i(x_m^i)(t_1)$ and $P_n^i(x_n^i)(t_1)$ to the individual decomposed property instances $P_r^i(x_r^i)(t_3)$ and $P_s^i(x_s^i)(t_3)$.

Diagrammatically, we then have:



19. I want to emphasize here that what follows is to be construed only as a representation of the correct relationship between emergent property instances and property instances on lower levels when causal sequences are involved. It is not to be construed as an argument that in all cases where we have different levels of property instances, emergentism holds. Supervenience does have some restricted uses.

I note here that decomposition does not have to occur. The system might stay at the $(i + 1)$ st level while it produces further $(i + 1)$ -level effects. Nor is it necessary that P_k^{i+1} be an emergent property. When it is not, the identity on the right side of the diagram will not hold, and the lower two tiers on the right will be missing. Perhaps some $i + 1$ property instances are primitive in this way, but this is doubtful given what we know of the evolution of our universe.

One further source of concern exists and it relates directly to assumption (6') of the downward causation argument. Is it possible for $i + 1$ -level instances to directly produce other $i + 1$ -level instances without synthesizing them from lower level instances? These higher level instances are usually emergent, and so it might be thought that they must themselves be formed by fusion from lower level instances and not by direct action at the higher level. This concern fails to give sufficient credit to the ontological autonomy of emergent property instances. Recall that i -level instances no longer exist within $i + 1$ -level instances—the higher level instances act as property instance atoms even though they may, under the right circumstances, be decomposed into lower level instances. It is perfectly possible for an $i + 1$ -level instance to be directly transformed into a different $i + 1$ -level instance (often with the aid of other property instances) or to directly transform another, already existing, $i + 1$ -level property instance (again usually with the aid of other property instances.) Simply because the i -level instances no longer exist, they can play no role in this causal transformation.

We can now see what is wrong with premises (5') and (6'). (5') misrepresents the way in which emergent property instances are produced, for as we have seen, the relationship between the higher and lower levels is not one of supervenience. A last reply is available to supervenience advocates insisting on the need for relations between properties. If the emergent instance is produced by a causal interaction, they can insist that causes require laws and that the generality inherent in laws requires properties as well as instances. This is not something that a thoroughgoing ontological approach needs to accept. Singular causes can be taken as fundamental,²⁰ and whether or not causal laws (or their statements) can then be formulated in terms of relations between properties (predicates) depends upon the complexity of the part of the world involved. Sometimes they can be, often they cannot.

We have already seen the problem with (6'): it is false to say that the i -level property instances co-occur with the $(i + 1)$ st level property instance. The former no longer exist when they fuse to form the latter.

20. See Humphreys 1989, Section 25.

That is why the notation used here is not entirely adequate, but this is almost inescapable given that formal syntax is ordinarily compositional. $[P_i^*P_s](x_u^i) + (x_v^i)(t_2')$ is not 'composed of' $P_i(x_u^i)(t_3)$ and $P_s(x_v^i)(t_3)$: a physical process of decomposition (better 'defusion') is required to create the last two.

So, we have given a construal of what $i + 1$ -level emergent properties might be, shown how they can have causal properties that are new in the sense that they are not possessed by their i -level origins, and provided the appropriate sense in which the emergent properties are instantiated only because lower level properties are instantiated. But there is now no sense in which, as the exclusion argument claims, i -level property instances are overdetermined by a combination of previous i -level instances and $i + 1$ -level instances. Nor are the $i + 1$ -level instances always required to be brought into being by some simultaneous set of i -level property instances, as the downwards causation argument asserts. I believe that there are two reasons why a solution of this kind is not immediately obvious when reading these arguments. First, by representing the emergence base within the downwards causation argument as a single, unanalyzed property P^* , the fact that emergent property instances are formed by a fusion process on lower level instances cannot be accurately represented. Secondly, by representing the situation in terms of properties rather than property instances, supervenience seems to be forced upon us, when in fact a treatment purely in terms of instances is open to us.

We do, of course, lose the causal closure of i -level property instances. This is not something that should disturb us overmuch. If a picture like the one I have described is roughly correct, then it turns out that an undifferentiated commitment to 'physicalism' is too crude and that both the 'mental' and the 'physical' are made up of multilayered sets of strata, each level of which is emergent from and (probably) only arbitrarily separable from the layers beneath it.

To sum up what we have established: the claim that an $i + 1$ -level emergent property is instantiated only because its i -level emergence base is instantiated is wrong—the 'emergence base' is not the reason the emergent property is instantiated—it is the move from the i -level to the $i + 1$ -level by fusion that gives us emergence. Second, the problem of overdetermination that is lurking behind downwards causation is now less problematical. We may maintain that all i -level events are determined by i -level antecedents but often this will be by way of j -level intermediaries.

So, to conclude: we have seen that two intriguing and widely canvassed arguments against emergent properties do not succeed in establishing their conclusions. It is more important, though, to emphasize

that a robustly ontic attitude towards emergent properties, rather than the more common logical approaches, can give us a sense of what emergent features might be like. Most important of all, I think, is to stop thinking of these issues exclusively in terms of mental properties, and to look for examples in more basic sciences.

In addition, construing the arguments in terms of multiple layers of property instances reminds us of two things: that there are many, many levels of properties between the most fundamental physical level and the psychological, and that it is left unacceptably vague in much of the philosophical literature just what is meant by ‘physical’. Being reminded of the large variety of property levels below the psychological, some of which are arguably emergent, should at least make us aware of the need to be more explicit on that score, and that *some* of the mysteries surrounding the physical/mental cleavage are perhaps the result of an inappropriate dichotomy.

6. From Metaphysics to Physics and Back. Thus far, the discussion has been completely abstract. My intention has been simply to show how one sort of emergent feature can avoid various difficulties inherent in supervenience treatments. What we have thus shown is the possibility of property instances being emergent, free from the difficulties stemming from the exclusion and downwards causation arguments. Even if there were no actual examples of fusion, the account of emergence given here would be useful because it provides a coherent account of a particular kind of emergence that is devoid of the mysteries associated with earlier attempts to explicate the concept. There is, of course, the further question of whether our world contains examples of emergent property instances. The answer to this is a reasonably confident ‘yes’. I shall here only sketch the form that such examples take, referring interested readers to more detailed sources for a richer description.

It frequently has been noted that one of the distinctive features of quantum states is the inclusion of non-separable states for compound systems, the feature that Schrödinger called “quantum entanglements.”²¹ That is, the composite system can be in a pure state when the component systems are not, and the state of one component cannot be completely specified without reference to the state of the other component. Furthermore, the state of the compound system determines the

21. The discussion of nonseparability goes back at least to Schrödinger 1935. d’Espagnat 1965 is another early source and more recent discussions can be found in Teller 1986, Shimony 1987, French 1989, Healey 1991, and d’Espagnat 1995, among many others.

states of the constituents, but not vice versa.²² This last fact is exactly the reverse of what supervenience requires, which is that the states of the constituents of the system determine the state of the compound, but when the supervening properties are multiply realizable, the converse does not hold. I believe that the interactions which give rise to these entangled states lend themselves to the fusion treatment described in the earlier part of this paper, because the essentially relational interactions between the ‘constituents’ (which no longer can be separately individuated within the entangled pair) have exactly the features required for fusion. One might be hesitant to use quantum entanglements as an argument by themselves because of the notorious difficulties involved in providing a realist interpretation for the theory. But what seems to me to be a powerful argument in favor of the existence of these emergent features is that these quantum entanglements are the source of macroscopic phenomena that are directly observable. In particular, the phase transitions that give rise to superconductivity and superfluidity in helium are a direct result of nonseparable states.²³

It is a question of considerable interest whether, and to what extent, fusion occurs in other areas. It would be a mistake to speculate on such matters, because the existence of such interactions is a contingent matter, to be settled by scientific investigation. There is indeed, within that part of metaphysics that can be (partially) naturalized, an important but neglected principle: *certain metaphysical questions cannot be answered (yet) because we do not know enough*. On the basis of this principle, those readers who want an answer as to whether, for example, mental phenomena are emergent in the above sense will, I am afraid, have to be patient.

There is one other way in which fusion can occur, and it is neither a matter of speculation nor something directly amenable to empirical inquiry. To see it, one simply needs to be reminded of something that has been lost in the avalanche of logical reconstructions of causation and other concepts in this century. It is that singular causal interactions between property instances, construed realistically, provide ‘horizontal’ examples of the kind of novelty that has here and elsewhere been discussed in ‘vertical’ terms. By ‘construed realistically’, I mean ‘taken to be *sui generis* features of the world, the properties of which are fundamentally misrepresented by reductive analyses or (humean) supervenience treatments of causation’.²⁴ This is not the place to persuade

22. See, e.g., Beltrametti and Cassinelli 1981, 65–72.

23. See Shimony 1993, 221.

24. I focus on causal interactions here only because of their familiarity. Other kinds of interactions can, one assumes, produce genuine novelty.

readers of the benefits of the realist singular view,²⁵ so I shall simply note that the issues of ‘horizontal’ and ‘vertical’ novelty are connected, for the explaining away of the former, especially by supervenience, tends to gain its plausibility from a sparse ontology of spacetime points possessing a restricted set of primitive physical properties. If you believe, in contrast, that solid state physics (for example) is more than just advanced elementary particle physics, you will begin to ask how phenomena from the two fields interact. You should then be prepared to find that emergence may be complicated, but that it is neither mysterious nor uncommon.²⁶

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25. For a partial account, see Humphreys 1989, Section 25.

26. Since this paper was first drafted in 1991 I have realized that the term ‘fusion’ has a standard use in the mereological literature that is almost opposite to its use here. I believe that my use is better justified etymologically, but mereology was there first. The reader is hereby advised never to confuse the two uses.